

# Cell-to-Cell Communication Model Optimization

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## 1 Introduction

This is the refined version of the Cell-to-Cell Communication model, presenting various methodologies for optimizing its performance. It offers insights beyond the correlation model, including the expression production model.

## 2 Model

The model is characterized by the following equations:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{B} + \boldsymbol{\epsilon} \quad (1)$$

Where  $\mathbf{B} \sim MVN(0, G)$ , and  $\epsilon \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ .

To define the dimension of each term:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} z_1 & 0 & \dots & 0 \\ 0 & z_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & z_n \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

The conditional expectation of  $y_i$  given  $x_i$  is expressed as:

$$E(y_i|x_i) = \beta_0 + \beta_1 x_i + b_i x_i \quad (2)$$

The covariance matrix  $\Sigma$  of  $\mathbf{Y}$  can be represented using  $\mathbf{Z}$  and the covariance matrix  $\mathbf{G}$  of  $\mathbf{B}$ :

$$\Sigma = Cov(\mathbf{Y}) = Cov(\mathbf{Z}\mathbf{B}) + Cov(\boldsymbol{\epsilon}) = \mathbf{Z}\mathbf{G}\mathbf{Z} + \sigma^2 \mathbf{I}_n \quad (3)$$

Hence, the likelihood function can be formulated as:

$$L(\Sigma, \beta_0, \beta_1, B; x_1, \dots, x_n) = (2\pi\Sigma)^{-\frac{N}{2}} \exp\left(-\frac{1}{2\Sigma}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{B})^2\right) \quad (4)$$

The log-likelihood is given by:

$$l = \log(L) = -\frac{n}{2} \log(2\pi\Sigma) - \frac{1}{2}((\mathbf{Y} - \mathbf{X}\beta - \mathbf{Z}B)^T \Sigma^{-1}(\mathbf{Y} - \mathbf{X}\beta - \mathbf{Z}B)) \quad (5)$$

Considering the penalty terms:

Lasso penalty (encouraging sparsity):

$$R_1(B) = \lambda_1 \|\mathbf{B}\| \quad (6)$$

Laplacian quadratic penalty (encouraging smoothness among nearby locations/cells):

$$W = D - G, \text{ where } D = \text{diag}(d_1, d_2, \dots, d_n), d_i = \sum_{j \neq i} g_{ij}$$

$$R_2(B) = \lambda_2 \|\mathbf{B}W\mathbf{B}\| \quad (7)$$

Therefore, the loss function can be expressed as:

$$L = l + R_1 + R_2$$

In this scenario, given the inadequacy of occurrences in every group (location/cell), we assume that the covariance between cells  $\Sigma$  is known and can be assumed as the inverse of distance between cells.

### 3 Optimization

Two approaches are considered for optimizing the model:

1. Finding the minimum of  $L$  by its derivative: It is similar to the maximum likelihood estimator; therefore, the estimator may be biased. Non-negativity is achieved by applying NNLS (Non-negative Least Squares) to constrain  $\mathbf{B}$ . Both the Newton-Raphson algorithm and L-BFGS-B (Limited-memory Broyden-Fletcher-Goldfarb-Shanno) algorithm can be used in this process.

2. Calculating  $\hat{\beta}$  and  $\hat{\mathbf{B}}$  through iteration:

$$\frac{dL}{d\beta} = 0$$

$$\hat{\beta}_{ML} = (\mathbf{X}^T \Sigma^{-1} \mathbf{X})^{-1} (\mathbf{X}^T \Sigma^{-1} (\mathbf{Y} - \mathbf{Z}\hat{\mathbf{B}})) \quad (8)$$

$\hat{\mathbf{B}}$  is the best linear unbiased estimator (BLUE) of  $\mathbf{B}$ .

$$\hat{\mathbf{B}}_{BLUP} = G\mathbf{Z}(\Sigma + \lambda_1 I + \lambda_2 W)^{-1}(\mathbf{Y} - \mathbf{X}\beta) \quad (9)$$

To achieve non-negativity, the NNLS (Non-negative least squares) can be applied to estimate  $\hat{\mathbf{B}}$ . Then iteration can be applied until  $\hat{\mathbf{B}}$  converge.

### 3.1 Optimization Method

For the two optimization techniques: Minimization of the loss function and iteration involving both  $\beta$  and  $\hat{\mathbf{B}}$ , both with and without non-negativity constraints, are executed to compare the results.

It’s worth noting that the production model, commonly employed in cell-to-cell communication, offers valuable insights in addition to the correlation model. In the production model, communication scores are represented in two ways: binary and continuous. Both scoring methods indicate that when either the ligand or receptor expression level is zero, the communication score for that ligand-receptor pair is zero. Consequently, we exclude cell pairs where either the ligand or receptor expression level is zero.

## 4 Model Performance

### 4.1 Simulation

The performance of models with four optimization method on simulation data (Minimum L / Iteration / Minimum L + Non-negative Constraint / Iteration + Non-negative Constraint) is as follows.

The models exhibit varying performance with respect to the spatial parameter  $b$ . Based on the results from the simulation dataset, the model with Minimum L optimization displays the highest level of accuracy.

When examining the overall correlation of ligand-receptor pairs among all cells, assuming a true  $\beta$  value of 1, the estimations for  $\beta_1$  across different models are presented in the following table:

| Model                               |            | P-value |                            |
|-------------------------------------|------------|---------|----------------------------|
| True                                | 1          | -       |                            |
| Minimum L                           | 0.9932214  | P-value | $1.226512 \times 10^{-09}$ |
| Minimum L + Non-negative Constraint | -0.2070727 | P-value | $2.033061 \times 10^{-01}$ |
| Iteration                           | 0.7809678  | P-value | $1.704383 \times 10^{-07}$ |
| Iteration + Non-negative Constraint | 0.501880   | P-value | $1.385434 \times 10^{-05}$ |

Table 1: Estimates of  $\beta_1$  in Different Models

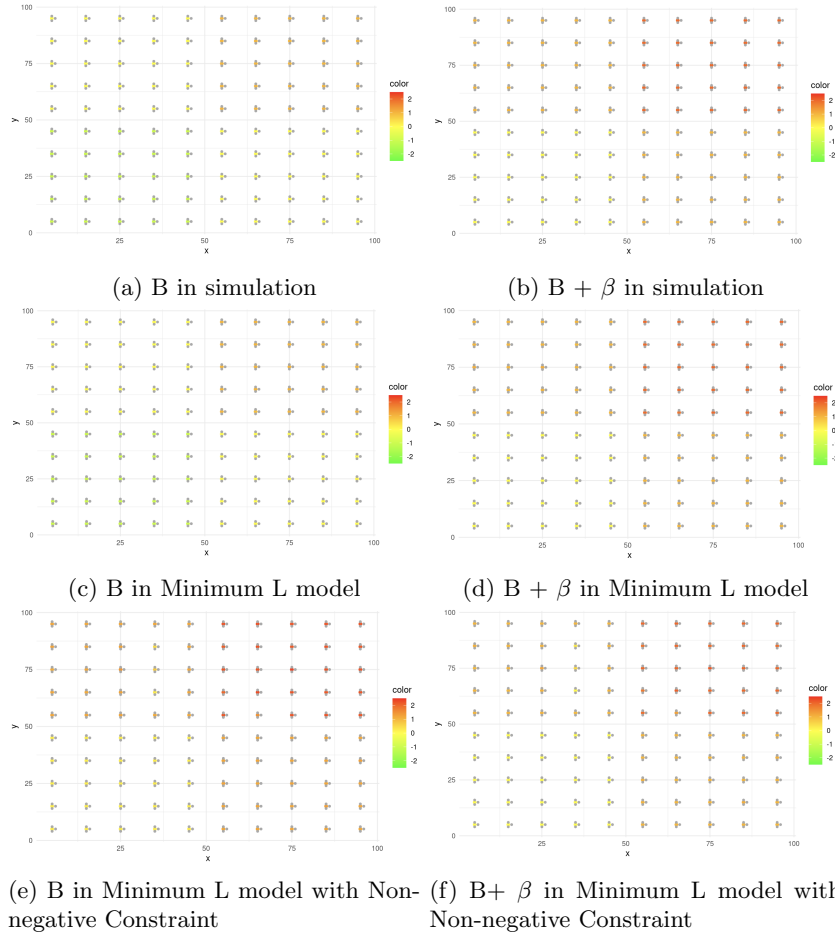


Figure 1: Comparison of Model Performances on Simulation Data (Page 1)

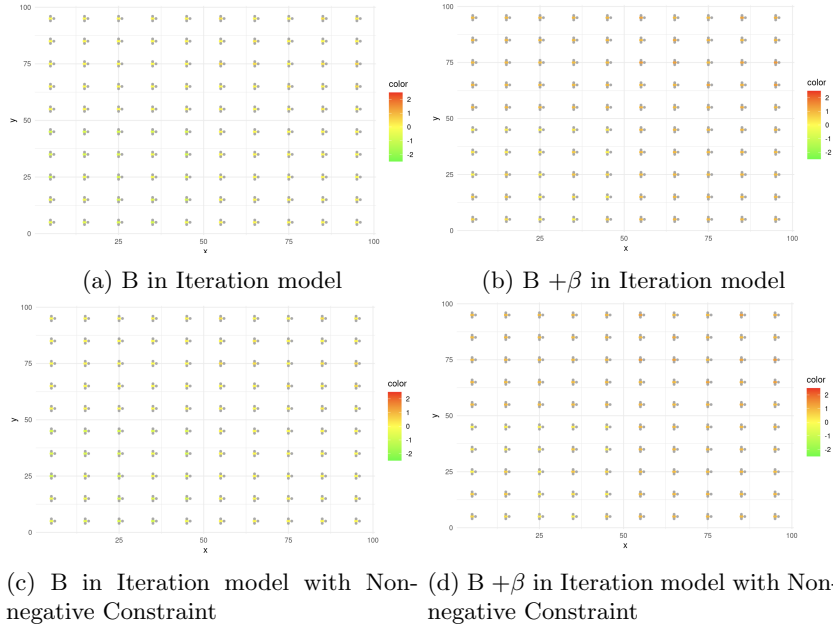
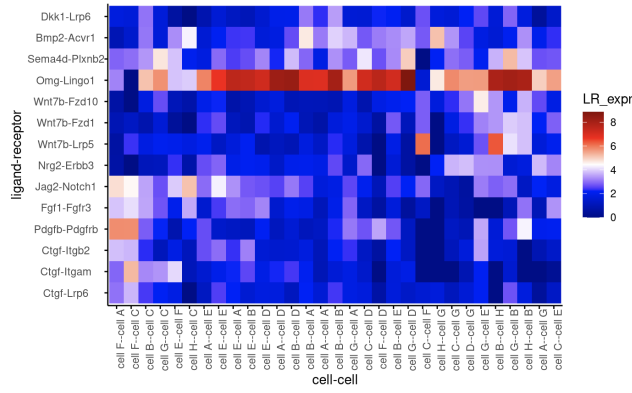


Figure 2: Comparison of Model Performances on Simulation Data (Page 2)

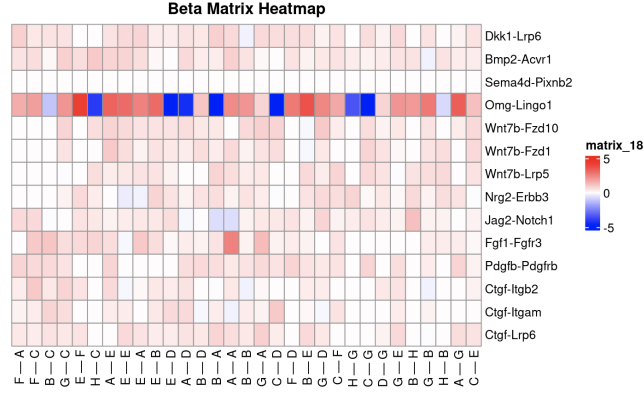
## 4.2 Benchmark

Here I have chosen two models that have shown better performance in terms of optimization (Minimum Loss and Iteration) from previous experiments in simulation, then benchmarked these two models using Giotto on the mini seqFISH dataset and seqFISH+Cortex dataset to evaluate their performance.

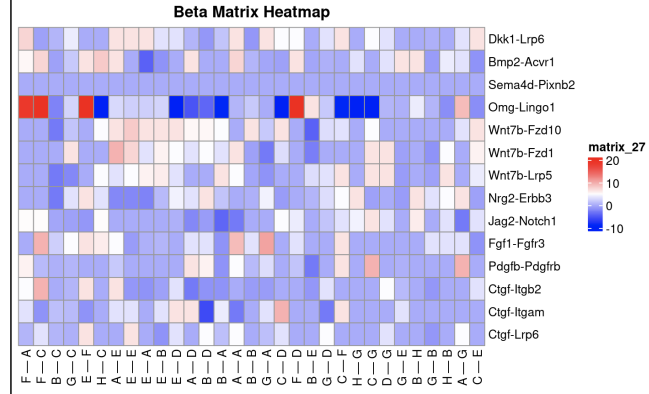
In the pictures, where the color represents the value of each LR-pair communication score for selected cell pairs, compared the performance of Giotto and current models. The cell annotations were generated either directly from the dataset or through clustering methods such as UMAP. These clustering methods can also be applied to the current model.



(a) Communication score on mini seqFISH by Giotto



(b) Communication score on mini seqFISH by Minimum L Optimization



(c) Communication score on mini seqFISH by Iteration Optimization

Figure 3: Comparison of Model of Communication score on mini seqFISH



### 4.3 Explanation

- **Why are some communication scores in this model negative?**

In the correlation model, a negative score indicates that there may not be a positive association between specific cell pairs concerning the overall relationship between lr pairs. However, when we consider the location-specific score  $\beta + B$ , some of these scores can still be positive.

In some other models, such as the expression production model, communication scores can only be positive. In this case, the strength of cell-cell communication, as well as the p-value, can be evaluated through permutations.

- **Why do the model performance and communication scores differ from the results of Giotto?**

Giotto utilizes the sum of the average expression to represent the communication score, ensuring non-negativity.

$$\overline{S(L, R, A, B)} = \frac{1}{n} \sum_{i=1}^n L_i + \frac{1}{m} \sum_{j=1}^m R_j;$$

Permutations are employed to generate the p-scores for communication scores. However, both models identify the same highly positive LR-pairs in specific cell pairs and highlight extreme values.

Notably, this model operates much faster than Giotto in terms of generating communication scores, without the need for permutations.